BiteSwipe

Product Overview & Instructions

CS 514 Team Project

November 2026

1 What is BiteSwipe?

BiteSwipe is a web app for finding somewhere to eat. It shows you San Diego restaurants one card at a time. You like the ones that look good and pass on the rest.

We built it because picking a restaurant with friends usually turns into twenty minutes of scrolling Yelp and Google Maps. Nobody decides anything. Swiping turns that into a quick yes or no. Every swipe gets saved, so the app remembers what you've already seen.

Everything lives in a normalized PostgreSQL schema: restaurants, cuisines, photos, opening hours, users and swipes. The feed is just a set of queries over those tables. We loaded about 550 real restaurants from Google Places and cleaned the data ourselves.

2 How it's built

There are three pieces. The React frontend shows the cards. The FastAPI backend handles logins and runs the queries. Postgres stores everything.

Part	**What we used**	**Where it runs**
Frontend	React + TypeScript (Vite, Tailwind)	Cloudflare Pages
Backend API	Python, FastAPI, SQLAlchemy	Heroku
Database	PostgreSQL (7 tables)	Neon
Login	Firebase Authentication (email + password)	Google
Restaurant photos	Image storage	Cloudflare R2

The main tables are `users`, `restaurants`, `cuisines`, `restaurant_cuisines`, `restaurant_photos`, `restaurant_hours` and `swipes`.

3 How to use it

The live app is at https://biteswipe-app.pages.dev/. It works in a phone or laptop browser.

1. **Make an account.** Open the app, click *Need an account?*, and register with an email and password. If you already have an account, just log in. Firebase handles the password. The first time you log in, the backend creates your row in `users`.
2. **Start swiping.** The home screen shows one restaurant card at a time with a photo, the name, the address and up to four cuisine tags.
   - Click **Like** (or press the right arrow key) if you'd go there.
   - Click **I'll Pass** (or press the left arrow key) to skip it.
3. **Keep going.** Every swipe is saved to the `swipes` table right away. You won't see the same restaurant twice.
4. **Log out.** Click **Log out** at the top of the page. Your swipes are tied to your account. They'll still be there next time.

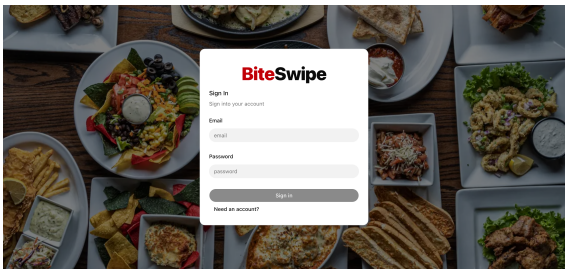


Figure 1: The login screen.

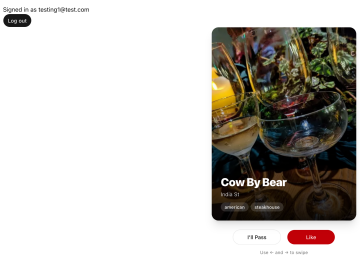


Figure 2: The card deck.

4 Running it locally

You only need this part to run BiteSwipe on your own machine. You'll need Python 3.12+, uv, Node 20+ and Docker.

1. **Clone the repo:** `git clone` https://github.com/spencer-barrett/CS514-BiteSwipe and `cd` into it.
2. **Start Postgres:** from the repo root, run `docker compose up -d`. This starts a local Postgres container.

3. **Set up the backend:** `cd apps/backend/api` and copy `.env.example` to `.env`. Fill in the Firebase credentials. The database URL already points at the container. Then run `uv sync`.

4. **Load the restaurant data:** `uv run alembic upgrade head`, then `uv run python -m biteswipe_api.scripts.seed data/clean`.

5. **Start the API:** `uv run uvicorn biteswipe_api.main:app -reload`. It runs at `http://localhost:8000`. You can try every endpoint at `/docs`.

6. **Start the frontend:** `cd apps/biteswipe-app` and copy `.env.example` to `.env`. Fill in the Firebase web config. Then run `npm install` and `npm run dev`, and open the URL Vite prints (usually `http://localhost:5173`).

5 Known limitations

- Restaurants come from a one-time snapshot of about 550 places in San Diego. New openings won't show up, and ratings won't update.
- The feed shows restaurants you haven't swiped on yet. The highest Google rating comes first. It doesn't use your likes to recommend places yet. That's the next thing we'd build on top of the `swipes` data.